1 Grammar Rules

Types and Contexts

$$
\begin{array}{llll}
\sigma, \tau & ::= & \mathbf{unit} \mid \mathbf{num} \mid \sigma \otimes \sigma \mid \alpha & \text{(Types)} \\
\alpha & ::= & m(\sigma) & \text{(Discrete Types)} \\
\Gamma & ::= & \emptyset \mid \Gamma, x :_r \sigma & \text{(Linear Typing Contexts)} \\
\Phi & ::= & \emptyset \mid \Phi, z : \alpha & \text{(Discrete Typing Contexts)}
\end{array}
$$

Expressions

$$
\begin{array}{lll}
e, f & ::= & x \mid z \mid () \mid !e \mid (e, f) \mid \mathbf{inl}\, e \mid \mathbf{inr}\, e \\
& \mid & \mathbf{let}\, x = e \,\mathbf{in}\, f \mid \mathbf{let}\, (x, y) = e \,\mathbf{in}\, f \mid \mathbf{dlet}\, z = e \,\mathbf{in}\, f \mid \mathbf{dlet}\, (x, y) = e \,\mathbf{in}\, f \\
& \mid & \mathbf{case}\, e' \text{ of } (\mathbf{inl}\, x.e \mid \mathbf{inr}\, y.f) \mid \mathbf{add}\, e\, f \mid \mathbf{sub}\, e\, f \mid \mathbf{mul}\, e\, f \mid \mathbf{dmul}\, e\, f \mid \mathbf{div}\, e\, f
\end{array}
$$

2 Typing Rules

Context Variable

$$\frac{}{\Phi \mid \Gamma, x :_r \sigma \vdash x : \sigma}\ \text{(Var)} \qquad \frac{}{\Phi, z : \alpha \mid \Gamma \vdash z : \alpha}\ \text{(DVar)}$$

Let Binding

$$\frac{\Phi \mid \Gamma \vdash e : \tau \qquad \Phi \mid \Delta, x :_r \tau \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let}\, x = e \,\mathbf{in}\, f : \sigma}\ \text{(Let)} \qquad \frac{\Phi \mid \Gamma \vdash e : \alpha \qquad \Phi, z : \alpha \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet}\, z = e \,\mathbf{in}\, f : \sigma}\ \text{(DLet)}$$

Tensor Destructor

$$\frac{\Phi \mid \Gamma \vdash e : \tau_1 \otimes \tau_2 \qquad \Phi \mid \Delta, x :_r \tau_1, y :_r \tau_2 \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let}\, (x, y) = e \,\mathbf{in}\, f : \sigma}\ (\otimes\mathrm{E}_\sigma)$$

$$\frac{\Phi \mid \Gamma \vdash e : \alpha_1 \otimes \alpha_2 \qquad \Phi, z_1 : \alpha_1, z_2 : \alpha_2 \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet}\, (z_1, z_2) = e \,\mathbf{in}\, f : \sigma}\ (\otimes\mathrm{E}_\alpha)$$

Bang Constructor

$$\frac{\Phi \mid \Gamma \vdash e : \sigma}{\Phi \mid \Gamma \vdash !e : m(\sigma)}\ \text{(Disc)}$$

Operation

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon + r_1} \mathbf{num}, y :_{\varepsilon + r_2} \mathbf{num} \vdash \{\mathbf{add}, \mathbf{sub}\}\, x\, y : \mathbf{num}}\ \text{(Add, Sub)}$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2 + r_1} \mathbf{num}, y :_{\varepsilon/2 + r_2} \mathbf{num} \vdash \mathbf{mul}\, x\, y : \mathbf{num}}\ \text{(Mul)}$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2 + r_1} \mathbf{num}, y :_{\varepsilon/2 + r_2} \mathbf{num} \vdash \mathbf{div}\, x\, y : \mathbf{num} + \mathbf{unit}}\ \text{(Div)}$$

$$\frac{}{\Phi, z : m(\mathbf{num}) \mid \Gamma, x :_{\varepsilon + r} \mathbf{num} \vdash \mathbf{dmul}\, z\, x : \mathbf{num}}\ \text{(DMul)}$$

References

[1] Ariel E. Kellison, Laura Zielinski, David Bindel, and Justin Hsu. Bean: A language for backward error analysis. *Proceedings of the ACM on Programming Languages*, 9(PLDI):1838–1862, 2025.